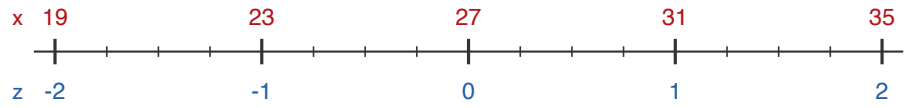
Answers: Z-Values on a Number Line

The mean is $\mu = 27$ and the standard deviation is $\sigma = 4$.

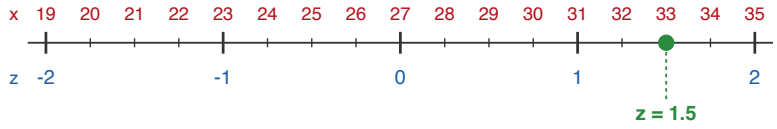
a) Fill in the x-values for each z-value:



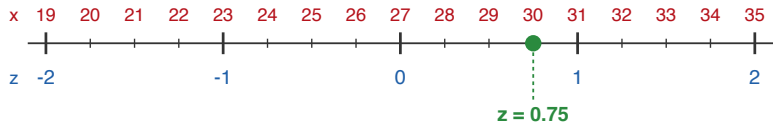
$\mu = 27$ is at $z = 0$. Each step of one σ adds or subtracts 4: $27 + 4 = 31$, $31 + 4 = 35$, $27 - 4 = 23$, $23 - 4 = 19$.

b) Find the z-score of each x-value below, using $z = \dfrac{x - \mu}{\sigma}$. Then mark x with a dot on its number line.

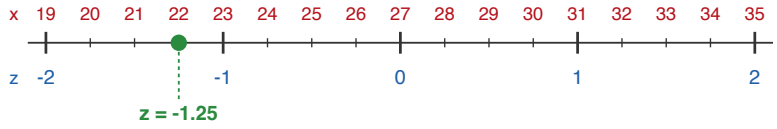
$x = 33$: $\quad z = \dfrac{33 - 27}{4} = \dfrac{6}{4} = 1.5$



$x = 30$: $\quad z = \dfrac{30 - 27}{4} = \dfrac{3}{4} = 0.75$



$x = 22$: $\quad z = \dfrac{22 - 27}{4} = \dfrac{-5}{4} = -1.25$



$x = 20$: $\quad z = \dfrac{20 - 27}{4} = \dfrac{-7}{4} = -1.75$

